

Holger Pirk

HOLGER PIRK – PERFORMANCE NUTRITION & FUELLING PLAN

Prepared by
Session date

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HOLGER PIRK – PERFORMANCE NUTRITION & FUELLING PLAN

1. Daily Energy & Macronutrient Targets

These targets are based on Holger's body weight (**79 kg**), current training load, and the need to support recovery, performance, and adequate energy availability.

Day Type	Training Load	Energy Target	Carbohydrate Target	Protein Target
Rest Day	Rest from running, light activity only	3000–3300 kcal	300–375g	125–140g
Easy Run Day	10km easy run + daily bodyweight work	3700–3900 kcal	395–475g	125–140g
Tempo / Speed Day	12km quality run + daily bodyweight work	3900–4150 kcal	475–630g	125–145g
Long Run Day	30–35km long run + daily bodyweight work	4500–4800 kcal	550–710g	130–145g

Main priorities

- Increase carbohydrate availability
- Fuel before and during key runs
- Recover properly after quality and long sessions
- Stop leaving too much intake until the evening
- Keep protein steady, but do not let it replace carbohydrate

2. Around-Run Fuelling Targets

A. Pre-Run Fuel

Easy run (<60–75 mins)

Target:

- **15–30g carbohydrate** if running early
- If running later, ensure the previous meal or snack contains carbohydrate

Tempo / speed session

Target:

- **80–120g carbohydrate** 2–3 hours before
- Optional extra **15–30g carbohydrate** 15–20 minutes before

Long run

Target:

- **100–150g carbohydrate** 2–4 hours before
- Optional extra **20–30g carbohydrate** 15–20 minutes before

B. During-Run Fuel

Easy run

Target:

- Usually not essential under 60–75 mins
- Water or electrolytes as needed
- Optional **15–30g carb** if starting under-fuelled

Tempo / speed session

Target:

- **30–45g carbohydrate per hour** if the session lasts >75 mins including warm-up/cool-down

Long run

Target:

- Build toward **60–80g carbohydrate per hour**
- Current average intake is about **30g/hour**, which is too low for marathon preparation

Fluids and sodium

- **2.6L/day baseline fluids**
 - **+ 500–750ml fluid per hour of training**
 - **1000–1500mg sodium across longer sessions**, depending on heat and sweat loss
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C. Post-Run Recovery

Easy run

Target:

- **40–60g carbohydrate + 20–25g protein** if appetite is low
- Or a normal balanced meal soon after

Tempo / speed / long run

Within **30–60 minutes** aim for:

- **80–95g carbohydrate**
- **24–30g protein**

This is especially important after:

- Wednesday quality session
- Thursday run if still carrying fatigue
- Sunday long run

3. Pre-Run Food Options

Easy Run Pre-Run Options (15–30g carbs)

Option	Food	Portion	Carbs
1	Banana	1 medium (120g)	25g
2	Dates	3 Medjool dates (60g)	30g
3	Toast + honey	1 slice toast (40g) + 10g honey	22g
4	Rice cakes + jam	2 rice cakes + 15g jam	25g
5	Apple juice	250ml	26g

Tempo / Speed Pre-Run Options (80–120g carbs)

Option	Food	Portion	Carbs	Protein
1	Oats, milk, banana, honey	80g oats + 300ml milk + 1 banana + 20g honey	95g	18g
2	Bagels, jam, skyr	2 bagels (180g) + 30g jam + 200g skyr	100g	22g
3	Cereal, milk, toast, jam	80g cereal + 300ml milk + 2 slices toast + 20g jam	95g	18g

4	Pancakes, syrup, fruit, yoghurt	3 pancakes + 30g maple syrup + 100g berries + 150g yoghurt	85–90g	15–18g
5	Rice pudding, banana, honey	300g rice pudding + 1 banana + 15g honey	85–90g	12–15g

Long Run Pre-Run Options (100–150g carbs)

Option	Food	Portion	Carbs	Protein
1	Large porridge	100g oats + 300ml milk + 1 banana + 20g honey	120g	20g
2	Bagels, banana, honey	2 bagels + 1 banana + 15g honey	120g	18g
3	Toast, jam, juice, yoghurt	3 slices toast + 40g jam + 250ml juice + 150g yoghurt	125–130g	18g
4	Pancakes, syrup, banana, yoghurt	4 pancakes + 40g syrup + 1 banana + 150g yoghurt	130–140g	20g
5	Cereal, milk, toast, jam	100g cereal + 300ml milk + 2 slices toast + 30g jam	135–140g	20g

15–20 Minute Top-Up Before Harder Runs (20–30g carbs)

Option	Food	Portion	Carbs
1	Banana	1 small	20g
2	Carb drink	300ml	20–25g
3	Energy chews	1 serving	20–25g
4	Dates	2 large	20g
5	Cereal bar	1 small bar	20–25g

4. During-Run Fuel Options (Practical for Running – No Gels)

Tempo / Speed Session: 30–45g carbohydrate per hour

Use simple, portable options that can fit in a shorts pocket, running belt, or soft flask.

Option	Food / Drink	Portion	Carbs/hour
1	Carbohydrate drink in soft flask	500ml	30g
2	Energy chews	1 pack / serving	30–40g
3	Carbohydrate drink + energy chews	300ml drink + ½–1 pack chews	35–45g

4	Chews + small carton of sports drink	1 serving chews + 250–300ml drink	40–45g
5	Homemade soft dates cut in half	4 small dates	28–32g

Long Run: 60–80g carbohydrate per hour

These are more realistic options for carrying on a run and using consistently.

Option	Food / Drink	Portion	Carbs/hour
1	Carbohydrate drink only	750ml soft flask / bottle	60g
2	Carbohydrate drink + energy chews	500ml drink + 1 serving chews	55–60g
3	Carbohydrate drink + chews + 2 dates	500ml drink + ½ serving chews + 2 dates	60–65g
4	Carbohydrate drink + small banana	500ml drink + 1 small banana	50–55g
5	Carbohydrate drink + 1 small oat-based running chew bar	500ml drink + 1 small bar	60g
6	Two soft flasks of carb drink	2 x 375ml flasks across the hour	60g
7	Carbohydrate drink + energy chews + extra sips as needed	500–750ml drink + 1 serving chews	60–75g

Best practical choices for Holger

These are likely the easiest options to use consistently:

Best Option	Why it works
750ml carbohydrate drink	Simplest option, easiest to tolerate, no chewing needed
500ml carbohydrate drink + energy chews	Very practical and easy to dose every 15–20 mins
500ml carbohydrate drink + 2–4 dates	Real-food option, portable if wrapped or stored in a small bag
500ml carbohydrate drink + small banana	Useful for steadier long runs if tolerated
Two soft flasks with carb mix	Easy to carry in vest/belt and sip gradually

5. Post-Run Recovery Options

Easy Run Recovery (40–60g carbs + 20–25g protein)

Option	Food	Portion	Carbs	Protein
1	Porridge + whey	60g oats + 250ml milk + 1 scoop whey	45–50g	28g
2	Toast + jam + quark	2 slices toast + 20g jam + 200g quark	45–50g	25g
3	Smoothie	1 banana + 250ml milk + 30g oats + 1 scoop whey	45–50g	28g
4	Cereal + milk + yoghurt	50g cereal + 250ml milk + 150g yoghurt	50g	20–25g
5	Rice pudding + skyr	200g rice pudding + 150g skyr	45–50g	22–25g

Tempo / Long Run Recovery (80–95g carbs + 24–30g protein)

Option	Food	Portion	Carbs	Protein
1	Oats + whey + banana + honey	80g oats + 1 scoop whey + 1 banana + 15g honey	85–95g	28–32g
2	Bagels + jam + quark	2 bagels + 30g jam + 200g quark	90–100g	28g
3	Smoothie	1 banana + 40g oats + 300ml milk + 1 scoop whey + 3 dates	85–95g	28–30g
4	Cereal + milk + fruit + whey	90g cereal + 300ml milk + 1 banana + 1 scoop whey	85–95g	30g
5	Rice pudding + yoghurt + honey	300g rice pudding + 150g yoghurt + 15g honey	85–95g	24–26g
6	Toast + jam + recovery shake	3 slices toast + 30g jam + 1 banana + 1 whey shake	85–95g	25–30g
7	Pretzels + smoothie + yoghurt	60g pretzels + fruit smoothie + 150g yoghurt	80–90g	22–28g

6.

A. Breakfast

Current	Issue	Improved version
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2 eggs + 200g tomatoes + 100g spinach + 60g groat porridge with 240g milk	Very good quality, but often too low in total carbohydrate for training load	Keep same base, but on training days increase to 80–100g oats , add 1 banana , and add 15–20g honey
2 eggs + veg + porridge	Too fibrous before hard sessions if eaten close to run	Before tempo/long runs, reduce veg volume and focus more on porridge + banana + toast + honey
Porridge alone before harder runs	May still be too small	Aim for 100–150g total carbs before long run using oats + milk + banana + honey + toast

Breakfast upgrades

Option	Improved breakfast	Carbs	Protein
1	100g oats + 300ml milk + 1 banana + 20g honey	120g	20g
2	80g oats + 300ml milk + 1 banana + 15g honey + 2 eggs	95g	28g
3	2 bagels + 30g jam + 200g skyr	100g	22g
4	100g cereal + 300ml milk + 2 slices toast + 20g jam	115g	18g
5	4 pancakes + 40g syrup + 1 banana + 150g yoghurt	130–140g	20g

B. Pre-Evening Run Snack

Current	Issue	Improved version
Dates before evening workout	Good start, but often too small	Pair with a clearer carb source
Leftover porridge	Fine, but not always enough	Standardise the portion
Quark alone	Too protein-heavy before running	Add carbohydrate alongside

Pre-run snack upgrades

Option	Snack	Carbs	Protein
1	3 dates + 1 banana	50g	1g
2	2 slices toast + 20g jam	40–45g	6g
3	200g leftover porridge + 10g honey	35–40g	6–8g
4	1 cereal bar + 1 banana	45–50g	3–5g

5	150g yoghurt + 1 banana + 20g honey	45–50g	10g
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C. During Long Runs

Current	Issue	Improved version
1 OTE bar per hour	Too low for marathon prep	Move toward 60–80g carbohydrate per hour
Mostly solid fuelling	Harder to hit higher carb targets	Use carb drink as the base
Average ~30g carb/hour	Too low for the demands of his training	Progress gradually over 3–4 weeks

Long-run fuelling upgrades

Option	During-run strategy	Carbs/hour
1	750ml carb drink	60g
2	500ml carb drink + 3 dates	60g
3	500ml carb drink + 30g oat bar	60g
4	500ml carb drink + 40g salted pretzels	60g
5	750ml carb drink + 1 small banana	75–80g

D. Post-Run Recovery

Current	Issue	Improved version
Silken tofu + fruit	Good quality, but often not enough carbohydrate	Add oats, toast, cereal, juice, or honey
Leftover porridge + whey	Better, but sometimes still too small after key sessions	Increase to full recovery target
Protein emphasis is good	Carbohydrate is still the main gap	Prioritise carbohydrate quantity after hard runs

5 exact post-run upgrades

Option	Recovery meal	Carbs	Protein
1	80g oats + whey + banana + 15g honey	85–95g	28–32g
2	2 bagels + 30g jam + 200g quark	90–100g	28g
3	90g cereal + 300ml milk + 1 banana + whey	85–95g	30g
4	300g rice pudding + 150g yoghurt + 15g honey	85–95g	24–26g

5	3 slices toast + 30g jam + 1 banana + whey shake	85–95g	25–30g
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E. Lunch & Dinner

Current	Issue	Improved version
Leftovers from dinner	Nutritious, but carbohydrate amount may vary too much	Make the carb portion more deliberate
Meals based on tofu, tempeh, lentils, veg	Good quality, but can become protein/fibre-heavy rather than carb-led	Add a defined grain or potato portion to each meal
Large intake in evening	Too much catch-up later in day	Bring more intake earlier through lunch/snacks

5 exact balanced lunch/dinner structures

Option	Meal	Carbs	Protein
1	300g cooked rice + 150g tofu + 200g veg	90–100g	25–30g
2	400g potatoes + 150g lentils + 200g veg	80–90g	25–30g
3	250g cooked barley + 150g tempeh + 200g veg	75–85g	28–32g
4	120g dry pasta + 150g lentil bolognese	85–95g	25–30g
5	250g cooked quinoa + 150g tofu + 1 bread roll + veg	80–90g	25–30g

F. Snack Improvements

Current	Issue	Improved version
Quark	Good protein, but often too low in carbs alone	Add granola, fruit, cereal or toast
Dates	Fine around training, but not enough by themselves for recovery	Pair with protein or more carbohydrate
Leftover porridge	Good option if portion is enough	Standardise the portion

5 exact snack options

Option	Snack	Carbs	Protein
1	200g quark + 40g granola + 100g berries	40–45g	20g
2	2 slices toast + 25g jam + 150g yoghurt	45–50g	15g

3	Smoothie: 1 banana + 40g oats + 250ml milk + whey	60–70g	25g
4	50g pretzels + 150g yoghurt	40–45g	12g
5	60g cereal + 250ml milk	50g	15g

7. Daily Timing Structure

Easy Run Day

Time	Target
06:30	15–30g carbs pre-run
07:00	Run
08:00–08:30	60–80g carbs + 25g protein
11:00	Snack: 30–40g carbs + 15–20g protein
13:30	Lunch: 80–100g carbs + 25–30g protein
16:30	Snack: 30–50g carbs + 10–15g protein
19:30	Dinner: 80–100g carbs + 25–30g protein
21:30	Optional pre-bed snack: 20–40g carbs + 10–20g protein

Tempo / Speed Day

Time	Target
06:00	80–120g carbs pre-run
07:30–09:00	30–45g carbs/hour during
Immediately after	80–95g carbs + 25–30g protein
11:30	Snack: 40–60g carbs + 15g protein
14:00	Lunch: 90–120g carbs + 25–30g protein
17:00	Snack: 30–50g carbs
20:00	Dinner: 90–120g carbs + 25–30g protein
21:30	Optional pre-bed carbs

Long Run Day

Time	Target
05:30–06:00	100–150g carbs pre-run

15 mins before	20–30g carbs top-up
During run	60–80g carbs/hour + 500–750ml fluid/hour
Within 30 mins	80–100g carbs + 25–30g protein
13:00	Lunch: 100–130g carbs + 25–30g protein
16:00	Snack: 50–70g carbs
19:30	Dinner: 100–130g carbs + 25–30g protein
21:30	Pre-bed carbs

8. Key Take-Home Points

1. **Carbohydrate is the main performance limiter here, not protein**
2. **Eat more before and after training, not just in the evening**
3. **Long-run fuelling must build from 30g/hour toward 60–80g/hour**
4. **Recovery after quality and long sessions should be structured, not optional**
5. **Use carb drink + easy foods rather than gels**
6. **Eat every 3–4 hours to improve recovery and reduce energy dips**